AutoGreen: Smart Traffic Signal Automation for Ambulance and Emergency Vehicles

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***Abstract*— Traffic jams at road intersections regularly delay ambulances, putting patients' lives at serious risk. To address this problem, this paper proposes AutoGreen, a reliable and straightforward RF-based traffic signal control system that operates without using GPS, GSM, or internet connectivity. In this system, a 433 MHz RF transmitter is installed inside the ambulance, which sends signals to RF receivers connected to an ESP32 at traffic signals. The system determines the direction from which the ambulance is approaching based on the received signal. Once a valid signal is detected, the normal traffic light cycle is temporarily overridden, and the corresponding signal is turned green using a relay module, allowing the ambulance to pass through the intersection without stopping. AutoGreen is low-cost, quick to respond, and easy to implement, making it an effective solution for reducing ambulance wait time and improving emergency response.**

***Keywords*— Smart Traffic Control, Ambulance Priority System, RF-Based Communication, ESP32, Traffic Signal Automation, Green Corridor, Emergency Vehicle Clearance, Intelligent Transportation System.**

I. INTRODUCTION

Urban traffic jams often delay ambulances at intersections with fixed time signals, putting patient safety at risk. Although GPS, RFID, GSM and vision-based systems have been proposed, they are often costly, complex or need internet access, which limits their use. This paper presents AutoGreen, a quick and low cost ambulance priority system that does not need GPS or internet. It uses a 433 MHz RF transmitter in the ambulance and RF receivers with an ESP32 controller at traffic lights. When an ambulance gets close, the signal turns green automatically with a relay, allowing the ambulance to pass through without stopping.

II. RELATED WORK

A. A number of systems have been proposed to provide priority for emergency vehicles at signalized intersections. A RF transmitter in the ambulance communicates with a RF receiver at the traffic signal to switch the signal to green when the ambulance approaches. However, their prototype operates with a single RF link and does not distinguish between different approaches or lanes at the junction, limiting directional control and scalability. [1]

B. An advanced traffic clearance system for ambulances was developed using an RF-434 module, where the ambulance is equipped with an RF transmitter and the traffic signal is fitted with an RF receiver controller by an ATmega328P microcontroller. When the ambulance siren is activate, the transmitted RF signal triggers the controller to switch the corresponding traffic signal to green, and in some cases, the ambulance location is shared using a Bluetooth- and GPS-enabled smartphone. While this system effectively clears intersections for emergency vehicles, it relies on a single RF receiver per junction and assume line-of-sight proximity, which limits lane-wise direction detection and multi-directional coordination at complex intersections .[2]

C. A smart traffic clearance system for emergency vehicles was developed by combining RFID and wireless communication for traffic signal pre-emption. In this approach, RFID receivers are installed at distances of 1km and 750m before the traffic signal, so that when an ambulance carrying an RFID tag enters the detection zone, the signal is automatically turned green in advance to reduce congestion. The system also integrates GSM communication to transmit patient details to the hospital. However, this method mainly relies on RFID-based identification and does not explicitly address multi-lane or direction-wise ambulance detection at the intersection itself.[3]

D. An Arduino-based automatic traffic clearance system was developed using RFID and IR sensors, where traffic density is measured through IR sensors and the traffic signals are dynamically controlled based on vehicle flow. Emergency vehicle priority is provided using an RF transceiver and RFID tags, allowing ambulance to override normal signal operation. While this approach effectively combines density based traffic control with emergency prioritization, it treats the junction largely as a single approach and does not identify the exact arm or lane from which the ambulance is approaching.[3]

E. Several smart ambulance systems have been developed using IoT and cloud-based architectures to manage emergency traffic clearance and patient monitoring. In such systems, RFID is used for traffic signal pre-emption, while ZigBee and and IoT cloud platforms are employed to transmit patient health parameters and ambulance status to hospitals. A priority-based algorithm dynamically changes traffic lights to support emergency vehicle movement. Although this approach enables end-to-end integration between the ambulance, traffic control units, and hospitals, it depends on centralized infrastructure, continuous internet connectivity, and multiple wireless technologies beyond RF links.[4]

F. AI and vision-based intelligent traffic light systems have been proposed to improve traffic congestion management and road safety at intersections. These systems use deep learning and computer vision techniques to analyze CCTV video feeds, detect traffic congestion and accidents, dynamically adjust green signal timings, and send wireless alerts to vehicles while updating centralized monitoring dashboards for traffic authorities. Although such approaches are highly effective for overall traffic management, they require camera infrastructure, high computational resources such as GPUs or edge devices, reliable network connectivity, and high deployment costs. As a result, they are less suitable as simple and low-cost retrofit solutions focused solely on ambulance traffic pre-emption in resource constrained environments .[5]

III HARDWARE COMPONENTS

The proposed AutoGreen Smart Ambulance System consists of a set of embedded and wireless hardware modules that work together to provide automatic traffic clearance for emergency vehicles. The core hardware structure of the system is inspired by RF- and RFID-based ambulance clearance architectures discussed in earlier work.

A. ESP32 Microcontroller

ESP32 is used as the main control unit in the proposed system. It receives RF signals from the receivers and controls the traffic signal switching through the relay module. The ESP32 performs real-time signal processing and dynamically controls traffic lights based on ambulance detection. Due to its high processing capability, low power consumption, and multiple digital input/output pins, the ESP32 is well suited for real-time embedded traffic control applications.[6]

B. RF Transmitter (433 MHz)

A 433 MHz RF transmitter is installed inside the ambulance and is used to send an emergency alert signal towards the traffic junction. The transmitter is manually activated by the ambulance driver during emergency situations, allowing the system to initiate traffic signal priority only when required. Once activated, the transmitter continuously broadcasts a wireless signal that can be detected by receivers placed at the intersection. This approach enables fast, wireless and low-latency communication between the ambulance and the traffic control unit, ensuring timely signal switching for emergency clearance.[2]

C. RF Receivers (433 MHz)

Multiple 433 MHz RF receivers are installed at the traffic signal junction to detect the wireless emergency signal sent from the ambulance. When the RF alert signal is received, the receiver outputs a digital signal to the microcontroller for further processing. This enables the traffic control unit to respond immediately to the ambulance approach. In the AutoGreen system, four RF receivers are placed around the junction to determine the exact direction from which the ambulance is approaching, allowing only the corresponding traffic signal to be switched to green. This improves accuracy and prevents unnecessary signal changes in other directions. [2]

D. Omni-Directional Antennas

Each RF receiver is connected to an omni-directional antenna to enable wide-area signal reception at the traffic junction. These antennas allow RF signals to be detected from all directions, ensuring reliable ambulance detection regardless of the approach path. This is especially important in busy urban environments where traffic density and signal obstructions are common. Wide area RF reception improves detection reliability and supports timely traffic signal switching during emergency situations. [2]

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E. Relay Module

A relay module is used as the switching interface between the ESP32 microcontroller and the traffic signal system. Once the ESP32 identifies the direction from which the ambulance is approaching, it activates the corresponding relay to change the traffic signal from RED to GREEN. The relay provides electrical isolation between the low power controller and the high power traffic lights, enduring safe and reliable operation. This type of relay-based switching is commonly used in emergency traffic clearance systems due to its robustness and ability to handle high voltage loads.[6]

F. Traffic Signal Light

The traffic signal unit consists of REF, YELLOW and GREEN indicator lights, representing a real-world four way traffic intersection. When the ESP32 activates the relay module, the signal corresponding to the ambulance's approach direction is switched to GREEN, while the signals in remaining directions remain RED. This ensures safe and controlled passage for the ambulance without disturbing cross traffic. The use of direction-wise traffic signal control improves efficiency and reduces unnecessary signal switching.[6]

G. Power Supply Unit

A regulated DC power supply is used to provide stable voltage to the ESP32 microcontroller, RF receivers and relay module. Proper power regulation helps prevent voltage fluctuations that could affect signal detection or control logic. A stable power source ensures continuous and reliable real-time operation of the system, which is critical for emergency traffic control applications.[2] .

IV WORKING METHODOLOGY

Step 1 – System Initialization at the Junction

At each four-way traffic junction, the ESP32 controller, four 433 MHz RF receivers, relay module, and traffic signal units are powered ON and initialized. During initialization, the ESP32 configures its GPIO pins to continuously monitor the outputs of the RF receivers and to control the relay modules connected to the red, yellow and green lights of each direction. This ensures that the system is ready to detect emergency signals and perform real-time traffic signal switching as soon as an ambulance approaches. Proper initialization of controller inputs and outputs is essential for reliable and uninterrupted traffic control operation. [1],[3]

Step 2 – Ambulance Transmitter Activation

When an emergency situation occurs, the ambulance driver manually activates the 433 MHz RF transmitter installed inside the vehicle. Once switched ON, the transmitter continuously broadcasts an RF alert signal along the ambulance route. This continuous transmission ensures that nearby traffic junctions can detect the approaching ambulance in advance and prepare for signal switching. Compared to RFID-based systems that operate only within a limited detection range, the RF-transmitter provides earlier and wider-area detection, which improves response time and reliability during high-speed or congested traffic conditions.[2],[3]

Step 3 –RF Signal Reception at the Junction

As the ambulance approaches the traffic junction, the RF alert signal transmitted from the vehicle is detected by the RF receiver positioned in the corresponding road direction. All four RF receivers continuously monitor for the incoming RF signal. The receiver facing the approaching ambulance detects the strongest signal and generates a digital HIGH output, which is sent to the associated ESP32 input pin. This enables the controller to recognize the presence of the ambulance and begin processing the emergency condition for traffic signal control.[2]

Step 4– Overriding Normal Signal Traffic

Once a valid RF signal and the ambulance approach direction are confirmed, the ESP32 temporarily overrides the normal timer-based traffic signal cycle and switches the junction into an ambulance priority mode. In this mode, the usual fixed-time or density-based signal sequence is paused so that immediate priority can be given to the ambulance. This allows the traffic signal to respond instantly instead of waiting for the normal cycle to complete, which is critical during emergency situations. Priority-based signal switching that suspends normal traffic operation and provides a green signal to the ambulance route plays an important role in reducing emergency response time at intersections.[6],[7]

Step 5 –Relay Operation and Green Corridor Creation

After the system enters ambulance priority mode, the ESP32 activates the relay corresponding to the detected ambulance direction, switching that particular lane's traffic signal to GREEN. At the same time, the relay controlling the remaining three directions are forced to RED, preventing cross traffic and ensuring a clear path for the ambulance. This relay-based switching allows the controller to safely interface with high-voltage traffic signal lamps and ensures reliable signal control during emergency situations. By keeping only the ambulance route green, the system effectively creates a temporary green corridor, allowing the ambulance to pass through the junction without interruption.[2],[6]

Step 6 –Continuous Monitoring during Ambulance Passage

While the ambulance crosses the intersection, the RF transmitter installed in the vehicle continues to broadcast the RF alert signal. The RF receiver at the junction continuously detects its signal, and the ESP32 maintains the green traffic signal for the ambulance route as long as the RF input remains active. This approach ensures that the ambulance is given uninterrupted passage through the junction without premature signal switching.

Once the ambulance moves away from the intersection and the RF signal strength drops below the detection threshold, the receiver output transitions to LOW, indicating that the ambulance has safely cleared the junction. Based on this change, the controller prepares to restore normal traffic operation. This method allows the system to automatically determine when the ambulance has passed and avoids unnecessary signal holding.[2],[6]

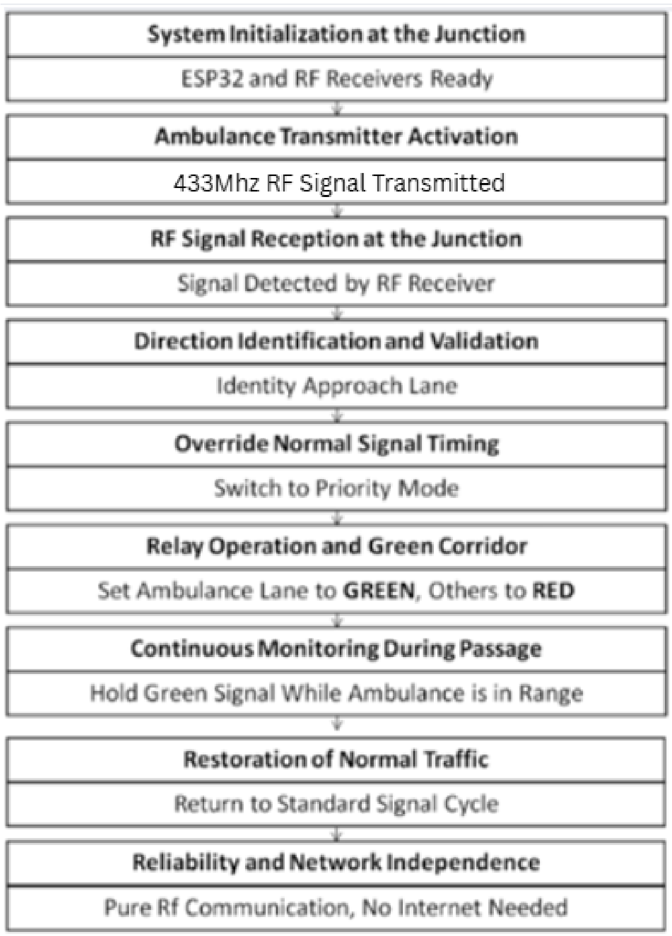
Step 7 – Restoration of Normal Traffic Operation

After confirming that the ambulance has safely left the junction, the ESP32 deactivates the ambulance-priority relay configuration and restores the standard red-yellow-green traffic signal sequence for all four directions based on the pre-programmed timing or density logic. This ensures that traffic priority is provided only for the required duration and the normal traffic flow is resumed without unnecessary delay. By performing this restoration locally at the junction, the system avoids dependence on cloud services or GPS-based coordination. This local control approach improves system readability and ensures continuous operation even in the absence of internet connectivity.[7]

Step 8 – Reliability and Independence from External Networks

By using pure RF communication between the ambulance and the traffic junction, the AutoGreen system avoids common issues such as GPS inaccuracy, central server dependency, and cloud failures that are often observed in traffic control systems relying on GPS- and cloud-based coordination. This standalone operation improves system reliability, especially in emergency situations where network availability cannot be guaranteed.

In addition, AutoGreen overcomes the limited detection range associated with RFID-based ambulance clearance systems by employing longer-range RF communication. This allows earlier detection of approaching ambulances and provides sufficient time for signal switching at busy intersections. By extending the concept of simple, fast, and low-cost RF-based ambulance clearance with direction-aware multi-receiver logic and ESP32-based control, AutoGreen enhances the effectiveness and practicality of RF-based emergency traffic management systems.[2],[3],[7]



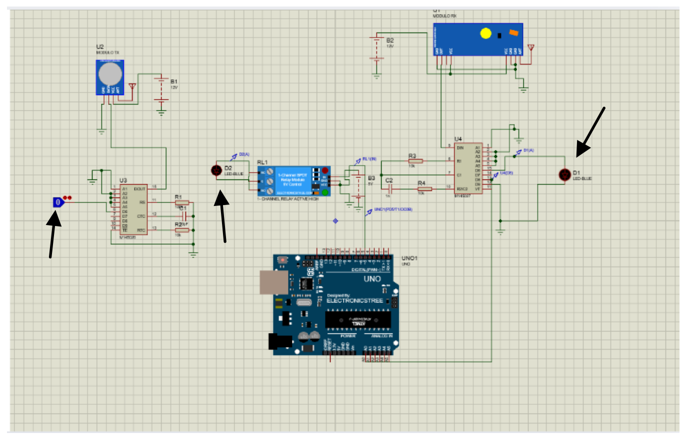
V. SIMULATION AND RESULTS

To verify the functional correctness of the proposed RF-based emergency signal control system, a complete hardware-level simulation was carried out using Proteus Design Suite. The simulation model consists of a 433 MHz RF Transmitter, a 433 MHz RF Receiver, an Arduino UNO microcontroller, a single-channel relay module, and two LED**s** used as visual indicators for traffic signal status. The RF transmitter represents the ambulance-side emergency trigger, while the RF receiver and Arduino represent the traffic junction control unit.

In the simulation setup, the RF transmitter is controlled by a digital switch, which acts as the emergency activation input. The DATA output pin of the RF receiver is connected to a digital input pin of the Arduino UNO. The Arduino processes this signal and controls the relay module through a digital output pin. The output of the relay is connected to two LEDs through suitable current-limiting resistors. These LEDs represent the traffic signal indicators. The simulation was performed under two operating conditions to validate system performance.[8]

A. Case 1: Switch = 0 (Normal Condition)

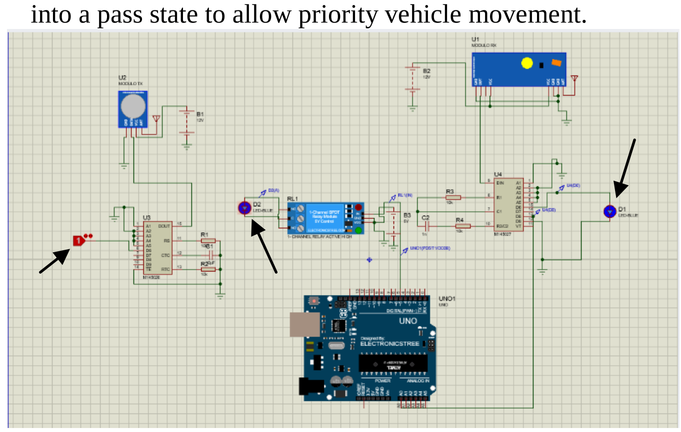
When the input switch is maintained at **logic level 0**, the RF transmitter remains in an inactive state and **no RF signal is transmitted**. As a result, the RF receiver output remains **LOW**. The Arduino continuously monitors this input and, since no valid signal is detected, it does not activate the relay module. Consequently, **both LEDs remain in the OFF state**. This condition represents the **normal traffic mode**, where no emergency vehicle is detected and the system stays in idle condition.



B. Case 2: Switch = 1 (Emergency Condition)

When the input switch is changed to **logic level 1**, the RF transmitter becomes active and begins transmitting a continuous RF signal. The RF receiver successfully detects this signal and its DATA output becomes **HIGH**. This digital signal is fed into the Arduino input pin. Upon detecting this HIGH signal, the Arduino executes the control logic and **energizes the relay module**. The relay contacts close and provide supply to both LED indicators. As a result, **both LEDs turn ON**, indicating that the emergency condition has been detected and traffic signal activation has occurred.

This operating state successfully emulates the **emergency signal clearance condition**, where traffic lights are forced into a pass state to allow priority vehicle movement.



C. Simulation Results and Verification

The Proteus simulation confirms the correct operation of the proposed RF-based control logic. The observed behavior strictly follows the expected design rule:

When **Switch = 0 → Both LEDs remain OFF**

When **Switch = 1 → Both LEDs turn ON**

This proves that:

The **RF communication link is functioning correctly**

The **Arduino correctly interprets RF receiver output**

The **relay module performs reliable switching**

The **LEDs accurately reflect the control output**

The real-time response of the system in the simulation indicates that the RF-based triggering mechanism is fast and reliable.

D. Significance of Simulation

The simulation demonstrates that the proposed system can successfully operate without the use of GPS, internet connectivity, or cloud servers. It proves the feasibility of using **simple RF-based wireless communication and microcontroller-based switching** for emergency signal control. Since the simulation uses standard hardware modules, the design can be directly implemented in real-world traffic signal systems by replacing LEDs with high-power traffic lamps and scaling the relay circuitry accordingly.

VI. IMPACT OF THE SYSTEM

The proposed AutoGreen Smart Ambulance Traffic Clearance System makes a meaningful difference when compared to conventional and IoT-based traffic control systems. One of its biggest strengths is that it works completely without GPS or internet connectivity. Because of this, it avoids common problems like network delays, cloud failures, and poor signal coverage. As a result, the system becomes more reliable and can function smoothly even in areas where connectivity is weak or unavailable.

By using simple RF-based communication, AutoGreen can respond instantly when an ambulance approaches. This is extremely important during emergencies, where even a few seconds can save lives. The quick response helps ambulances move faster through intersections without unnecessary delays. Studies have also shown that RF-based systems provide faster response times and are cost-effective, making them ideal for real-time emergency applications [2], [4], [10].

Another important impact of AutoGreen is its simplicity and low cost. Unlike advanced AI- or camera-based traffic systems that require heavy processing, constant monitoring, and complex algorithms, this system uses basic components such as an RF transmitter, RF receivers, an ESP32 controller, and relay modules. This makes it easier to install, maintain, and operate, while also reducing energy consumption. Because of this, the system can be implemented more easily in many locations.

The system also creates a real-time green corridor for ambulances by automatically adjusting traffic signals. This helps reduce waiting time at intersections and allows emergency vehicles to pass quickly. Faster movement of ambulances can directly contribute to saving lives, as delays in traffic are minimized [9], [10].

In busy urban areas, AutoGreen further improves efficiency by detecting ambulances from a longer distance using RF signals. This early detection allows the system to prepare traffic signals in advance, improving accuracy and timing. Additionally, by using multiple RF receivers, the system can identify the exact direction from which the ambulance is coming, ensuring that only the correct lane is given priority [9].

Overall, AutoGreen focuses on being simple, fast, and affordable. It is especially useful for small- and medium-sized cities, as well as in situations where advanced infrastructure or internet connectivity is not available. Because of these advantages, it stands out as a practical and effective solution for improving emergency response systems [10].

VII FUTURE INTEGRATION

A. City-Wide Multi-Intersection Coordination

In the future, the AutoGreen Smart Ambulance System can be expanded from controlling a single traffic signal to managing multiple traffic junctions across an entire city. At present, the system gives priority to an ambulance at one intersection. By connecting nearby junctions, a continuous green path can be created, allowing the ambulance to move smoothly without stopping at every signal. This kind of coordination helps reduce overall delay and ensures faster ambulance movement, especially during heavy traffic conditions.[6]

B.Integration with IoT and Hospital Communication Systems

Another important future improvement is integrating AutoGreen with **IoT-based hospital communication systems**. With this integration, information such as the ambulance location, expected arrival time, and patient condition can be sent directly to nearby hospitals. This allows doctors and medical staff to prepare in advance before the ambulance arrives. Combining smart traffic control with real-time medical information can reduce treatment delays and help improve patient survival during emergencies.[4]

C. AI-Based Traffic Awareness and Incident Detection

In the long term, AutoGreen can be enhanced by adding **AI-based traffic monitoring and incident detection**. AI-enabled camera systems can automatically detect traffic jams, accident, and unusual road conditions in real time. By using this information, traffic signals can adjust themselves not only for ambulances but also based on AI and RF-based control can make traffic management smarter, safer, and more effective during emergency situations.[5]

VIII CONCLUSION

An RF-based traffic signal clearance system aimed at reducing ambulance delays at urban intersections. The system uses a **433 MHz RF transmitter installed inside the ambulance**, which is **manually activated activated by the driver during an emergency**, When the ambulance approaches a signal and the transmitter is activated, the system identifies the direction of approach and switches the corresponding traffic light to green using a relay module. This ensures smooth and uninterrupted ambulance movement without relying on GPS, internet connectivity, or cloud infrastructure.

The proposed system was validated through hardware design and Proteus-based simulation, which confirmed reliable RF communication, correct controller operation, and effective relay-based signal switching under both normal and emergency conditions. Compared to existing GPS. RFIS and AI-based solutions, AutoGreen provides a **simple, low-cost and direction-aware approach** that is practical for real-world deployment, especially in small and medium-sized cities. With future enhancements such as multi-intersection coordination and IoT integration, AutoGreen has strong potential to further improve emergency response efficiency and help save lives.

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